

Properties of System

- causality(因果性)
- memoryless (无记忆性)
- linearity (线性性)
- time invariance (时不变性)
- stability (稳定性)

causality

- its output depends only on current and past inputs.
- a continuous-time signal $x : \mathbb{R} \rightarrow A$
- $x|_{t \leq \tau}$ represent a function called the restriction in time
$$x|_{t \leq \tau}(t) = x(t)$$
- if x is an input to a system, then $x|_{t \leq \tau}$ is the “current and past inputs” at time τ .

causality

- This system is causal if for all $x_1, x_2 \in X$ and $\tau \in \mathbb{R}$,

$$x_1|_{t \leq \tau} = x_2|_{t \leq \tau} \Rightarrow S(x_1)|_{t \leq \tau} = S(x_2)|_{t \leq \tau}$$

- All systems we have considered so far are causal

strictly causal

- for all $x_1, x_2 \in X$ and $\tau \in \mathbb{R}$,
$$x_1|_{t < \tau} = x_2|_{t < \tau} \Rightarrow S(x_1)|_{t \leq \tau} = S(x_2)|_{t \leq \tau}$$
- for two possible inputs x_1 and x_2 that are identical up to (not including) time τ , outputs are identical up to (including) time τ .
 - The Integrator actor: strictly causal;
 - the adder: not strictly causal, but causal;
- Strictly causal actors are useful for constructing **feedback** systems.

Memoryless Systems

- $S : X \rightarrow Y$, where $X = A^{\mathbb{R}}$ and $Y = B^{\mathbb{R}}$
- exist a function $f : A \rightarrow B$ such that for all $x \in X$,

$$(S(x))(t) = f(x(t))$$

- output is decided by input at time t .
- example:the adder is the memoryless system, but the Integrator is not.
- if a system is strictly causal and memoryless then its output is constant for all inputs (?)

Linearity and Time Invariance

- superposition property (叠加性质) :

for a system $S: X \rightarrow Y$,

$$\forall x_1, x_2 \in X \text{ and } \forall a, b \in \mathbb{R},$$

$$S(ax_1 + bx_2) = aS(x_1) + bS(x_2).$$

- A system is linear if it satisfies the superposition property

- example

$$\dot{\theta}_y(t) = \dot{\theta}_y(0) + \frac{1}{I_{yy}} \int_0^t T_y(\tau) d\tau.$$

$$\forall t \in \mathbb{R}, \quad y_2(t) = i + \int_0^t x_2(\tau) d\tau.$$

- cascade of any two linear actors is linear.
- the Scale actor, the adder are always linear

Linearity and Time Invariance(LTI)

- delay actor: $D_\tau : X \rightarrow Y$, where X and Y are sets of continuous-time signals, be defined by

$$\forall x \in X \text{ and } \forall t \in \mathbb{R}, (D_\tau(x))(t) = x(t - \tau).$$

τ is a parameter of the delay actor

- A system $S : X \rightarrow Y$ is time invariant if

$$\forall x \in X \text{ and } \forall \tau \in \mathbb{R}, S(D_\tau(x)) = D_\tau(S(x)).$$

- linear time-invariant system (LTI): linear and time-invariant

- A major objective in modeling physical dynamics is to choose an LTI model

Stability

- **bounded-input bounded-output stable (BIBO stable or just stable)** if the output signal is bounded for all input signals that are bounded.

- example

$$\dot{\theta}_y(t) = \dot{\theta}_y(0) + \frac{1}{I_{yy}} \int_0^t T_y(\tau) d\tau.$$

- ♦ Let the input be $T_y = u$, where u is the **unit step**

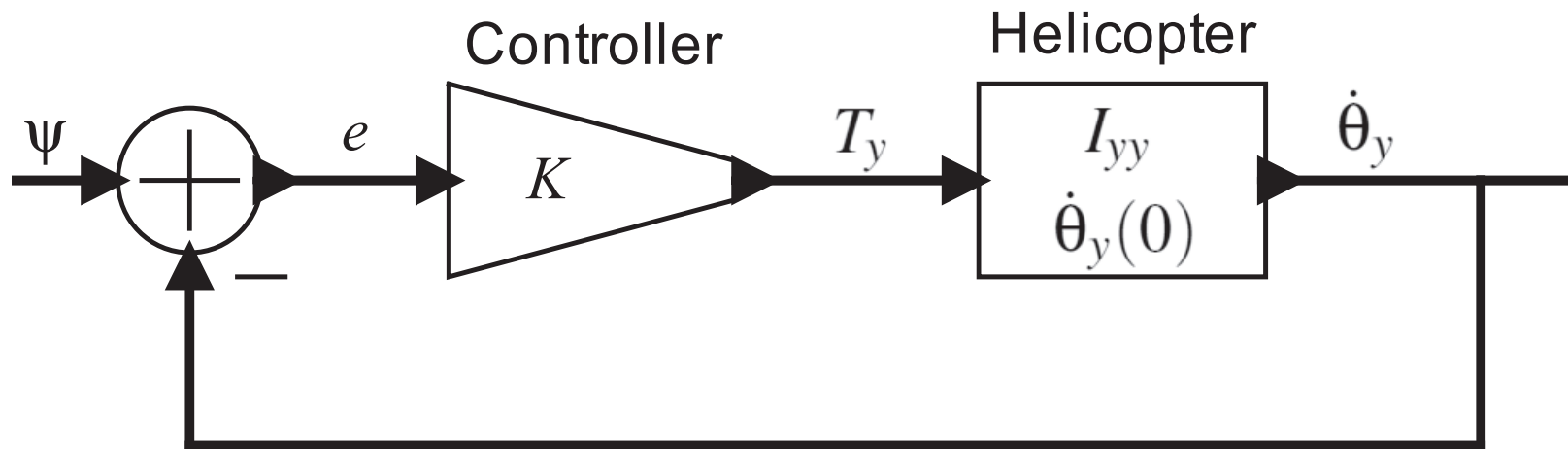
$$\forall t \in \mathbb{R}, \quad u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

- ♦ This input is clearly bounded.

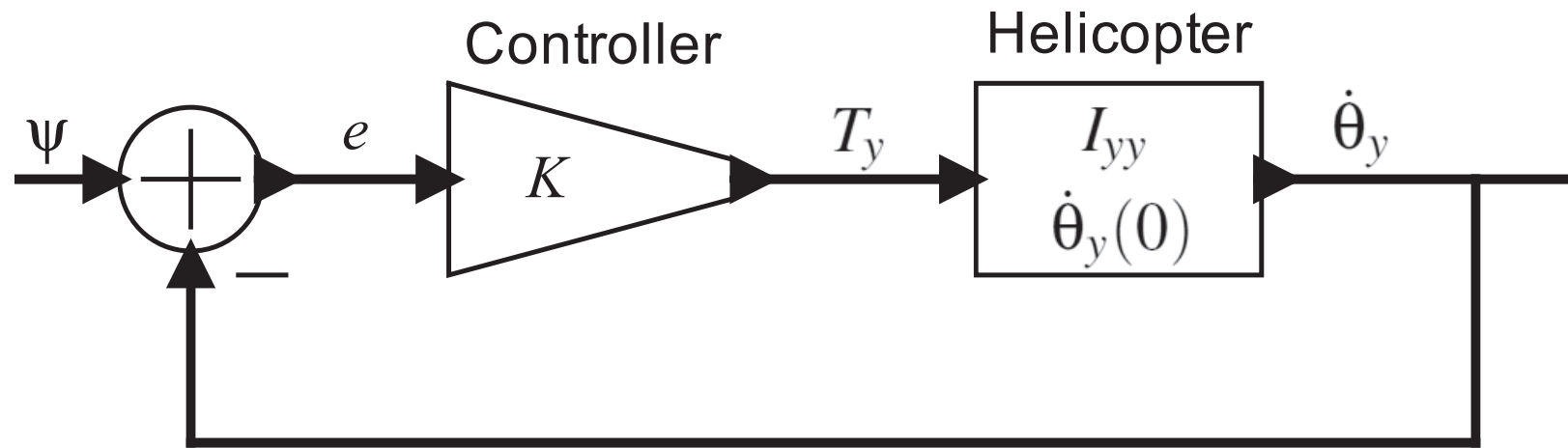
- ♦ the output grows without bound.

2.4 Feedback Control

- A system with **feedback** has directed cycles, where an output from an actor is fed back to affect an input of the same actor.

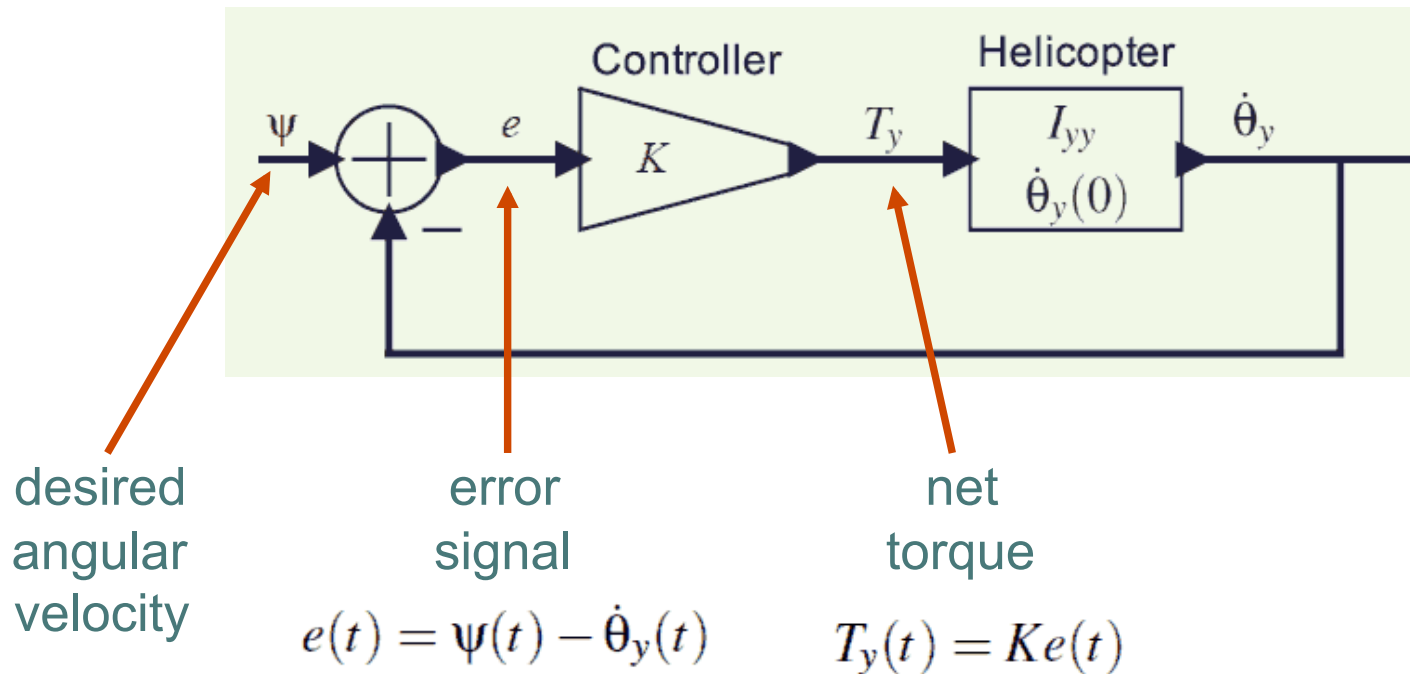


Feedback Control



- comparison desired behavior (ψ) and actual behavior ($\dot{\theta}_y$)
- the corresponding correction signal (T_y) should compensate to reduce future error
- at least one strictly causal system

Proportional controller

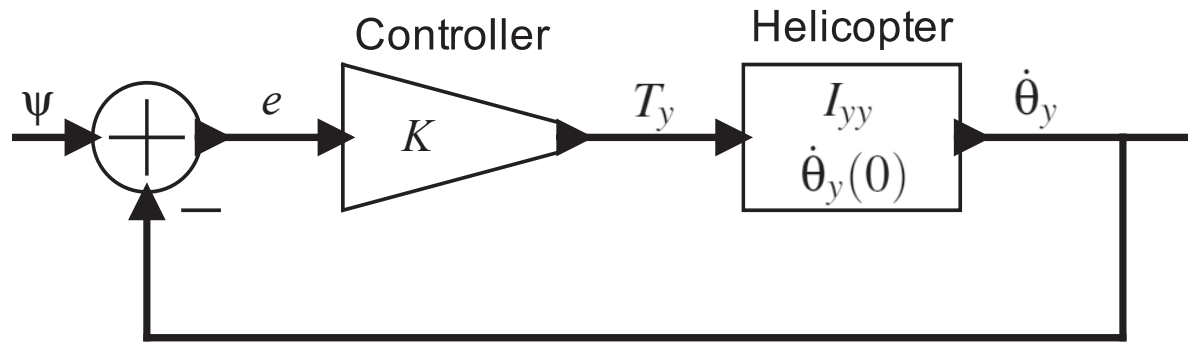


$$\begin{aligned}\dot{\theta}_y(t) &= \dot{\theta}_y(0) + \frac{1}{I_{yy}} \int_0^t T_y(\tau) d\tau \\ &= \dot{\theta}_y(0) + \frac{K}{I_{yy}} \int_0^t (\psi(\tau) - \dot{\theta}_y(\tau)) d\tau,\end{aligned}$$

Note that the angular velocity appears on both sides, so this equation is not trivial to solve.

- control signal is proportional to the error.

example1:Proportional control feedback



$$e(t) = \psi(t) - \dot{\theta}_y(t), \quad T_y(t) = Ke(t).$$

$$\dot{\theta}_y(t) = \dot{\theta}_y(0) + \frac{K}{I_{yy}} \int_0^t (\psi(\tau) - \dot{\theta}_y(\tau)) d\tau,$$

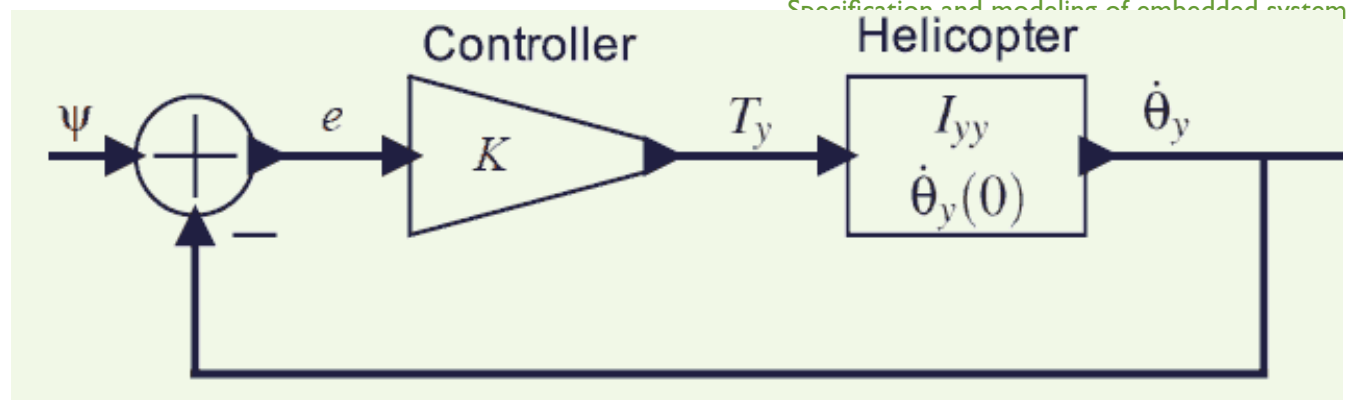
assuming that $\psi(t) = 0$

$$\dot{\theta}_y(t) = \dot{\theta}_y(0) - \frac{K}{I_{yy}} \int_0^t \dot{\theta}_y(\tau) d\tau.$$

the solution is

$$\dot{\theta}_y(t) = \dot{\theta}_y(0) e^{-Kt/I_{yy}} u(t).$$

example2



$$\dot{\theta}_y(t) = \dot{\theta}_y(0) + \frac{K}{I_{yy}} \int_0^t (\psi(\tau) - \dot{\theta}_y(\tau)) d\tau$$

Assume that helicopter is initially at rest,

$$\dot{\theta}(0) = 0,$$

and that the desired signal is

$$\psi(t) = au(t)$$

for some constant a .

By calculus (see notes), the solution is

$$\dot{\theta}_y(t) = au(t)(1 - e^{-Kt/I_{yy}})$$

$$\begin{aligned} &= \frac{K}{I_{yy}} \int_0^t (\psi(\tau) - \dot{\theta}_y(\tau)) d\tau \\ &= \frac{K}{I_{yy}} \int_0^t a d\tau - \frac{K}{I_{yy}} \int_0^t \dot{\theta}_y(\tau) d\tau \\ &= \frac{Kat}{I_{yy}} - \frac{K}{I_{yy}} \int_0^t \dot{\theta}_y(\tau) d\tau. \end{aligned}$$

example3

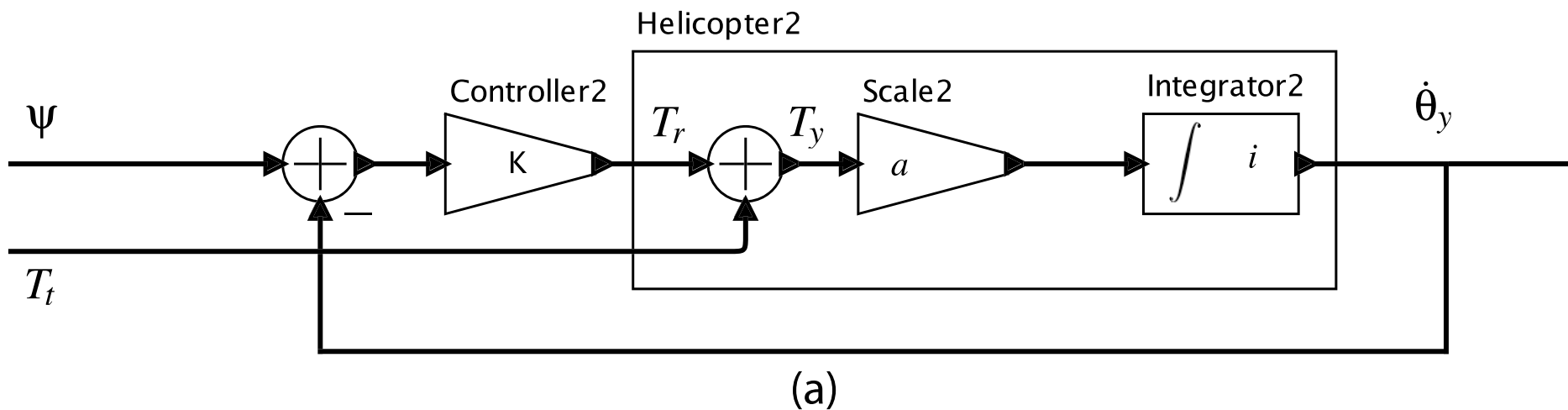
- the net torque T_y is the sum of the torque T_t due to the friction of the top rotor and the torque T_r due to the tail rotor,

$$\forall t \in \mathbb{R}, T_y(t) = T_t(t) + T_r(t).$$

where T_t is determined by the rotation to an altitude.

- design a control system that controls T_r

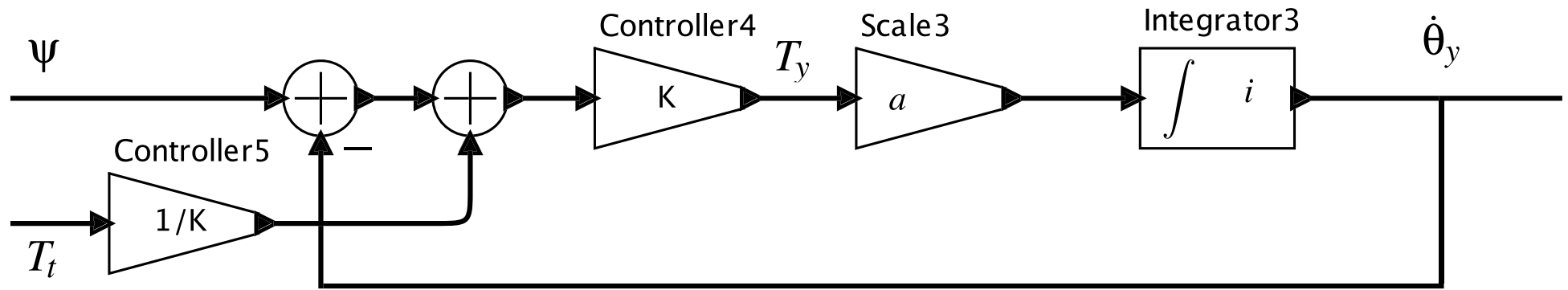
example3 (cont')



- Helicopter model with separately controlled torques for the top and tail rotors.

$$\dot{\theta}(0) = 0, \quad T_t = bu(t)$$

example3 (cont')

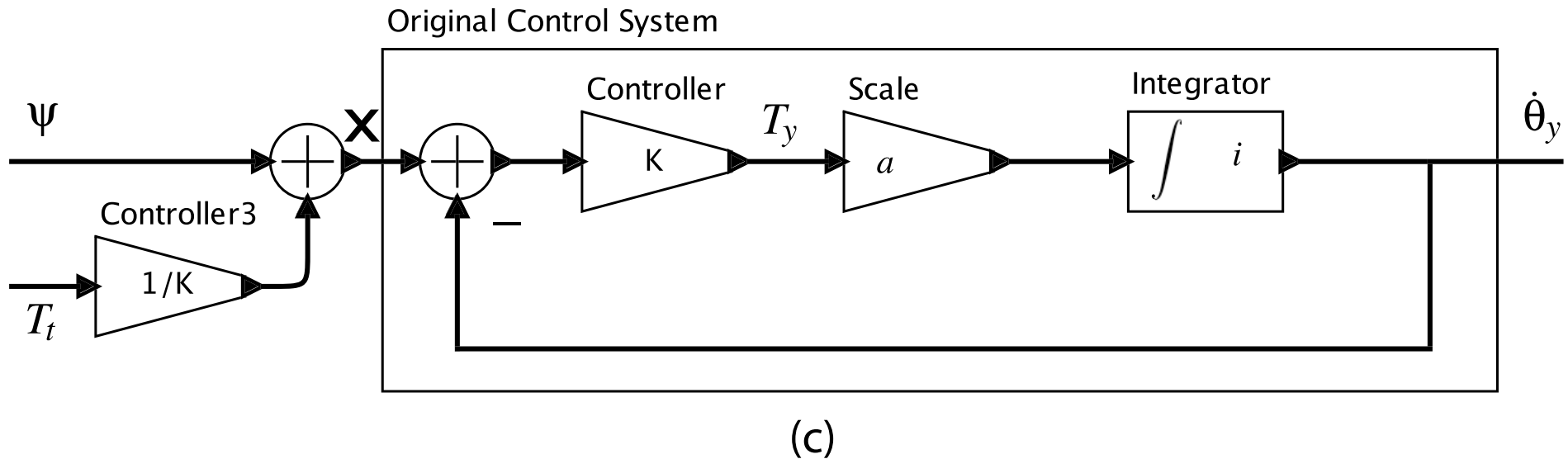


(b)

- Transformation to an equivalent model (assuming $K > 0$).
- reason is

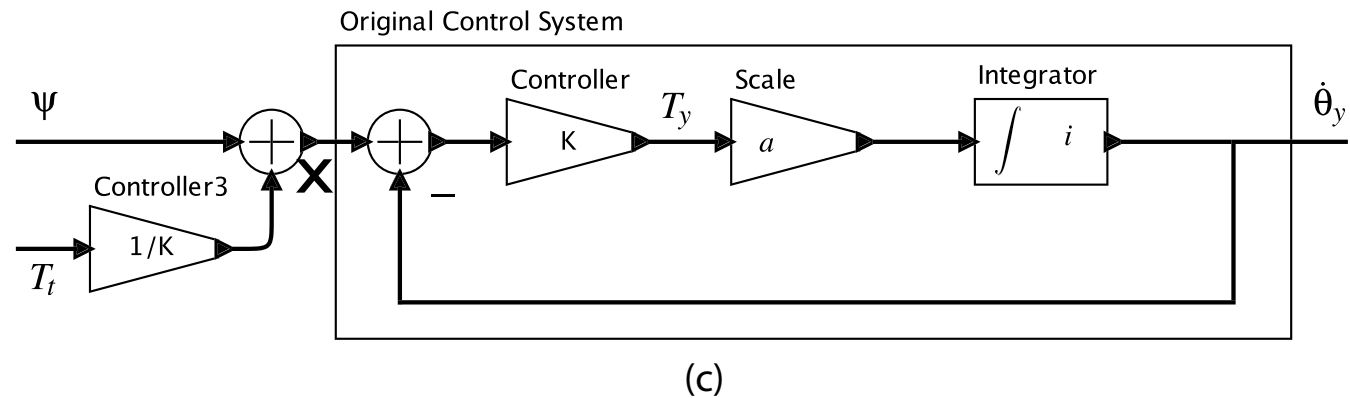
$$K a_1 + a_2 = K(a_1 + a_2/K).$$

example3 (cont')



- addition is commutative.
- Further transformation to an equivalent model that we can use to understand the behavior of the controller.

example3 (cont')



Assuming $\psi(t)=0$.

Then the input to the original control system will be

$$x(t) = \psi(t) + T_t(t)/K = (b/K)u(t).$$

the solution is

$$\dot{\theta}_y(t) = (b/K)u(t)(1 - e^{-Kt/I_{yy}}).$$

summary

- methods of modeling physical dynamics
 - ordinary differential equation
 - actor model---software modeling
- properties of system
 - linearity
 - time invariance
 - stability
- feedback control system